

Copula-Based Mean-Reverting Markov Processes: A Corrected and Self-Contained Theoretical Framework

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Abstract

This document replaces the theoretical part of `meanreversion_article_main.tex`. It separates exact finite-horizon conditional-mean identities from infinitesimal generator statements, formulates the absolutely continuous Markov copulas needed in the empirical analysis directly through their densities, and states the domain, support, boundary, integrability, and monotonicity conditions that are needed to turn formal PDE solutions into valid Markov models with observable transition densities. The corrected framework retains the valid OU, CIR, monotone-transform, mixed-copula reset, and nonlocal bond-pricing results. It treats the two Gaussian–Gamma calculations as necessary-condition diagnostics, restricts the quadratic CIR transform to a finite horizon, identifies the admissibility problem for Kummer branches, excludes the even OU transform from the one-dimensional monotone framework, and replaces the erroneous second CIR change of variables. The final section states exactly how each empirical specification relates to the corrected theory.

1 Probability space, supports, and notation

Work on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t \geq 0}, \mathbb{P})$ satisfying the usual conditions. Conditional laws given a point are always versions of regular conditional probability kernels; equalities involving such versions are asserted for the relevant marginal law almost everywhere unless continuity extends them to all interior states.

For a distribution function F , its generalized quantile is

$$Q(u) = F^{-1}(u) := \inf\{x \in \mathbb{R} : F(x) \geq u\}, \quad 0 < u < 1. \quad (1)$$

Quantiles are not assumed finite or differentiable at $u = 0, 1$. The support of the law at time t is denoted by $E_t = \text{supp}(F_t)$. A statement that a CDF is strictly increasing means strictly increasing on the interior of its support, not on all of $\mathbb{R}$. The symbol m denotes the target mean, $\bar{r}$ a base diffusion's long-run level, and s a Gamma scale. These objects were all denoted by θ in the original manuscript and must not be conflated.

2 Markov–copula foundations

2.1 Absolutely continuous Markov copulas

Let $I = (0, 1)$ and let Leb be Lebesgue probability measure on I . Set $\Delta = \{(s, t) : 0 \leq s < t\}$.

Assumption 2.1 (Absolutely continuous Markov–copula density evolution). *There is a jointly*

Borel-measurable function $c : \Delta \times I^2 \rightarrow [0, \infty)$; write $c_{s,t}(u, v) = c(s, t, u, v)$. For every $s < t$,

$$\begin{aligned} \int_I c_{s,t}(u, v) dv &= 1, & \text{for every } u \in I, \\ \int_I c_{s,t}(u, v) du &= 1, & \text{for Leb-a.e. } v \in I, \end{aligned} \quad (2)$$

$$c_{s,t}(u, v) = \int_I c_{s,r}(u, w) c_{r,t}(w, v) dw, \quad \begin{aligned} 0 \leq s < r < t, \quad u \in I, \\ \text{for Leb-a.e. } v \in I. \end{aligned} \quad (3)$$

For $s < t$, define

$$C_{s,t}(x, y) = \int_0^x \int_0^y c_{s,t}(u, v) dv du, \quad (x, y) \in [0, 1]^2. \quad (4)$$

On the diagonal set $C_{t,t}(x, y) = \min(x, y)$ and use the identity transition operator; no Lebesgue density $c_{t,t}$ is asserted.

Proposition 2.2 (Density consistency and the copula star product). *Under Assumption 2.1, (4) is a copula with density $c_{s,t}$. The operators*

$$(P_{s,t}g)(u) = \int_I g(v) c_{s,t}(u, v) dv, \quad P_{t,t} = I, \quad (5)$$

form a Markov evolution preserving the uniform law. Equivalently, the pair-copulas obey

$$C_{s,t} = C_{s,r} * C_{r,t}, \quad (C * D)(u, v) = \int_I \partial_2 C(u, w) \partial_1 D(w, v) dw. \quad (6)$$

For every $0 \leq t_0 < \dots < t_n$, the finite-dimensional density

$$f_{t_0, \dots, t_n}(u_0, \dots, u_n) = \prod_{j=1}^n c_{t_{j-1}, t_j}(u_{j-1}, u_j) \quad (7)$$

is projectively consistent. Hence these densities define a Markov process U with uniform marginals on the continuous time index. No sample-path continuity is implied by this construction alone.

Proof. Nonnegativity and the two identities in (2) give groundedness, two-increasingness, and uniform margins. Fubini's theorem turns (3) into both operator composition $P_{s,r}P_{r,t} = P_{s,t}$ and the star product (6). The first marginalization in (7) uses the column normalization, the last uses the row normalization, and removal of an intermediate time uses (3). Thus the family is projectively consistent. Since I is a standard Borel space, the Kolmogorov extension theorem supplies the stated process. $\square$

Remark 2.3. This article deliberately restricts attention to absolutely continuous pair-copulas because every copula used in the empirical likelihood has a density. This is a simplifying scope assumption, not a claim that all Markov copulas are absolutely continuous. Consistency (3), rather than the validity of isolated bivariate copulas at selected horizons, is what supports a continuous-time Markov model.

2.2 Uniformization and exact conditional-expectation identities

Assumption 2.4 (Regular marginal representation). *For every t , F_t is continuous and strictly increasing on E_t^c , Q_t is finite on I , and $\int_I |Q_t(u)| du < \infty$. The process U has the transition densities in Assumption 2.1, is adapted to $(\mathcal{F}_t)$, and is Markov with respect to this filtration. Moreover,*

$$X_t = Q_t(U_t) \quad \mathbb{P}\text{-a.s.} \quad (8)$$

Theorem 2.5 (Exact quantile–density criterion). *Under Assumption 2.4, for $s < t$,*

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = \int_I Q_t(v) c_{s,t}(U_s, v) dv \quad \mathbb{P}\text{-a.s.} \quad (9)$$

Consequently, the integrable process X is a martingale with respect to $(\mathcal{F}_t)$ if and only if

$$Q_s(u) = \int_I Q_t(v) c_{s,t}(u, v) dv \quad \text{for every } s < t \text{ and Lebesgue-a.e. } u. \quad (10)$$

Proof. The Markov property of U with respect to $(\mathcal{F}_t)$ and (8) give (9). Since $Q_s(U_s) = X_s$ and U_s is uniform, the martingale identity $\mathbb{E}[X_t \mid \mathcal{F}_s] = X_s$ is equivalent to (10) up to Lebesgue-null sets. $\square$

Proposition 2.6 (Observable transition density). *If, in addition, for every t , F_t is absolutely continuous with density $p_t > 0$ on E_t° , then a transition-density version of X is*

$$p_{s,t}^X(x, y) = c_{s,t}(F_s(x), F_t(y)) p_t(y), \quad x \in E_s^\circ, \quad y \in E_t^\circ, \quad (11)$$

with the conditional-law identity understood for $p_s(x) dx$ -a.e. x and Lebesgue-a.e. y . It satisfies

$$\begin{aligned} \int_{E_t^\circ} p_{s,t}^X(x, y) dy &= 1, \\ \int_{E_s^\circ} p_s(x) p_{s,t}^X(x, y) dx &= p_t(y), \\ \int_{E_r^\circ} p_{s,r}^X(x, z) p_{r,t}^X(z, y) dz &= p_{s,t}^X(x, y), \quad s < r < t, \end{aligned}$$

in the corresponding almost-everywhere sense.

Proof. Given $X_s = x$, the variable U_t has density $c_{s,t}(F_s(x), v)$. The change of variables $v = F_t(y)$ gives (11). Row normalization gives the first identity, column normalization gives marginal propagation, and the substitution $w = F_r(z)$ in (3) gives the physical-state Chapman–Kolmogorov identity. $\square$

2.3 Generators and the true-martingale qualification

For bounded continuous g , use the transition operators (5).

Definition 2.7 (Right and extended generators). *For a Feller evolution, the right generator at time t is*

$$\mathcal{A}_t g = \lim_{h \downarrow 0} \frac{P_{t,t+h} g - g}{h} \quad (12)$$

whenever the limit exists in the chosen Banach-space norm. An unbounded time-dependent quantile $q(t, u)$ is said to lie in the time–space extended generator domain if a localized Dynkin formula holds with drift $(\partial_t + \mathcal{A}_t)q$ and the required integrability is finite.

Theorem 2.8 (Generator martingale theorem). *Let $q(t, u) = Q_t(u)$ lie in the time–space extended generator domain. Suppose that, for every $T < \infty$, the stopped family*

$$\{q(\tau, U_\tau) : \tau \text{ is a stopping time with } 0 \leq \tau \leq T\}$$

is uniformly integrable, that is, $q(\cdot, U_\cdot)$ is of class–(D) on every finite horizon. Then $X_t = q(t, U_t)$ is a true martingale if and only if

$$(\partial_t + \mathcal{A}_t)q(t, u) = 0 \quad \text{for } dt \otimes \text{Leb}(du)\text{-a.e. } (t, u). \quad (13)$$

Without the class–(D) or an equivalent uniform-integrability condition, the zero-drift equation generally establishes only a local martingale.

Proof. The extended Dynkin formula makes $q(t, U_t) - q(0, U_0) - \int_0^t (\partial_s + \mathcal{A}_s)q(s, U_s) ds$ a local martingale. Equation (13) therefore gives a local martingale; class-(D) upgrades it to a true martingale. Conversely, the finite-variation part of a special-semimartingale decomposition is unique, so a true martingale in the stated domain has zero drift almost everywhere. The measure-theoretic details, including the passage from $dt \otimes \mathbb{P}$ to $dt \otimes \text{Leb}$, are given in Appendix A.4. $\square$

2.4 Gaussian and diffusion copula generators

Let Φ and ϕ be the standard Normal CDF and density.

Proposition 2.9 (Brownian and deterministic-time-change generators). *For the uniformized Brownian family and $0 < s < t$, the pair-copula is Gaussian with correlation $\rho_{s,t} = \sqrt{s/t} \in (0, 1)$ and density*

$$c_{s,t}^G(u, v) = \frac{\phi_{\rho_{s,t}}(\Phi^{-1}(u), \Phi^{-1}(v))}{\phi(\Phi^{-1}(u)) \phi(\Phi^{-1}(v))}, \quad (14)$$

where ϕ_ρ is the standard bivariate Normal density with correlation ρ . The raw standardized Brownian example is defined on $(0, \infty)$. For $t > 0$, its local generator on $C_c^2(I)$ is

$$\mathcal{A}_t g(u) = -\frac{z\phi(z)}{t} g'(u) + \frac{\phi(z)^2}{2t} g''(u), \quad z = \Phi^{-1}(u). \quad (15)$$

If Brownian time is replaced by a differentiable, strictly increasing h with $h(t) > 0$, replace $\rho_{s,t}$ in (14) by $\sqrt{h(s)/h(t)} \in (0, 1)$, and

$$\mathcal{A}_t^h g(u) = \frac{h'(t)}{h(t)} \left[-z\phi(z)g'(u) + \frac{1}{2}\phi(z)^2 g''(u) \right]. \quad (16)$$

In particular, $h(t) = e^{2at}$ with $a > 0$ gives

$$\mathcal{A}g(u) = a\phi(z)^2 g''(u) - 2az\phi(z)g'(u). \quad (17)$$

For both clocks, $\rho_{s,t} = \rho_{s,r}\rho_{r,t}$ for $s < r < t$; Gaussian conditioning therefore gives the density Chapman–Kolmogorov identity (3). A specification whose time index includes $t = 0$ must use a clock with $h(0) > 0$ rather than $B_t/\sqrt{t}$ at the origin.

Proof. For $Z_t = B_t/\sqrt{t}$, Itô's formula gives $dZ_t = -Z_t(2t)^{-1}dt + t^{-1/2}dB_t$. Applying Itô again to $U_t = \Phi(Z_t)$ yields (15). The deterministic time change multiplies the local clock by $h'(t)$ and evaluates the Brownian coefficient at $h(t)$, which gives (16). $\square$

Proposition 2.10 (Uniformized diffusion generator). *Let*

$$dR_t = b(t, R_t)dt + \sqrt{a(t, R_t)}dW_t \quad (18)$$

take values in an interval (ℓ, r) . Suppose that its CDF $F \in C^{1,2}$ in the interior, $p = \partial_x F > 0$, and a is locally C^1 in x . Write $F_t^R(x) = F(t, x)$ and $Q_t^R = (F_t^R)^{-1}$. Assume that, for every $s < t$, a jointly Borel transition-density family gives the entire regular conditional law

$$\mathbb{P}(R_t \in dy \mid R_s = x) = p_{s,t}^R(x, y) dy, \quad y \in (\ell, r),$$

with no additional boundary atom. Then its pair-copula has density

$$c_{s,t}(u, v) = \frac{p_{s,t}^R(Q_s^R(u), Q_t^R(v))}{p(t, Q_t^R(v))}. \quad (19)$$

If in addition the left-boundary probability flux

$$b(t, x)p(t, x) - \frac{1}{2}\partial_x\{a(t, x)p(t, x)\} \longrightarrow 0 \quad \text{as } x \downarrow \ell \quad (20)$$

vanishes, then the local generator of $U_t = F_t(R_t)$ is

$$\mathcal{A}_t g(u) = [a p_x + \frac{1}{2}a_x p]_{x=Q_t^R(u)} g'(u) + \frac{1}{2} [a p^2]_{x=Q_t^R(u)} g''(u). \quad (21)$$

Proof. The conditional change of variables $y = Q_t^R(v)$ gives (19). Its row normalization is the normalization of $p_{s,t}^R$; its column normalization follows from marginal propagation $p(t, y) = \int_{\ell}^r p(s, x) p_{s,t}^R(x, y) dx$. With $z = Q_r^R(w)$ and $dw = p(r, z) dz$, the physical transition-density Chapman–Kolmogorov identity gives (3). Apply the time-dependent Itô formula to $F_t^R(R_t)$ and use the integrated Fokker–Planck equation. Condition (20) removes the boundary constant. A final Itô application to $g(U_t)$ gives (21). $\square$

Remark 2.11. The preceding interior result covers OU directly. CIR is degenerate at zero and must be treated on $(0, \infty)$ with its nonnegative solution, interior density, and boundary behavior stated separately; a uniformly elliptic whole-line assumption cannot be invoked for CIR.

3 Correct notions of mean reversion

3.1 Local, conditional, and auxiliary notions

Definition 3.1 (Forward local drift). *For an integrable real-valued Markov process with transition kernel $P_{s,t}(x, dy)$, its forward local drift is*

$$b_X(t, x) = \lim_{h \downarrow 0} \frac{\int y P_{t,t+h}(x, dy) - x}{h}, \quad (22)$$

whenever a finite limit exists. The definition is made for a chosen regular kernel version and for F_t -almost every $x \in E_t$; a continuous version may be extended to all interior states.

Definition 3.2 (One-dimensional Markov mean reversion). *An integrable real-valued Markov process X is locally mean reverting toward m if*

- (i) $\lim_{t \rightarrow \infty} \mathbb{E}[X_t] = m$;
- (ii) *the forward drift (22) exists and*

$$(x - m)b_X(t, x) < 0 \quad (x \neq m), \quad b_X(t, m) = 0 \quad (23)$$

whenever the indicated states belong to the support. The quantity b_X is a drift, not a “reversion rate.”

Definition 3.3 (Strong conditional-mean reversion). *An integrable adapted process X has strong conditional-mean reversion toward m if a deterministic function $\mathcal{R}_{s,t} > 0$ exists such that*

$$\mathbb{E}[X_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(X_s - m), \quad 0 \leq s < t, \quad (24)$$

$$\mathcal{R}_{s,r}\mathcal{R}_{r,t} = \mathcal{R}_{s,t}, \quad \lim_{t \rightarrow \infty} \mathcal{R}_{s,t} = 0. \quad (25)$$

A principal case is $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u) du\}$ with continuous $\gamma(u) < 0$.

Definition 3.4 (Finite-horizon and directional classes). *For model diagnostics we use two weaker labels, neither of which is silently called global mean reversion.*

- (i) *A finite-horizon conditional-reversion model satisfies (24) and is a one-dimensional Markov model only on a stated window $0 \leq t \leq T$.*
- (ii) *A directional-reversion model has a drift pointing toward a threshold c , but c has not been proved equal to the limiting mean.*

An observed projection of a Markov vector is called a latent-state extension; it is not asserted to be a one-dimensional Markov process.

Proposition 3.5 (Strong reversion implies local reversion). *Suppose Definition 3.3 holds with $\mathcal{R}_{s,t} = \exp\{\int_s^t \gamma(u) du\}$, where γ is continuous and negative. Then*

$$b_X(t, x) = \gamma(t)(x - m). \quad (26)$$

If X_s is integrable, then $\mathbb{E}[X_t] \rightarrow m$, so X satisfies Definition 3.2.

Proof. Set $s = t$ and replace t in (24) by $t + h$; differentiation at $h = 0+$ gives (26). Taking expectations in (24) and using (25) gives the limiting mean. Since $\gamma(t) < 0$, the sign condition follows. $\square$

Remark 3.6. The converse is false in general: an instantaneous sign condition and an unconditional limiting mean do not by themselves determine every finite-horizon conditional mean. This is why the local and strong definitions are kept separate.

Proposition 3.7 (Martingale rescaling). *Let X be an integrable Markov process satisfying Definition 3.3. Then*

$$M_t = \frac{X_t - m}{\mathcal{R}_{0,t}} \quad (27)$$

is a Markov martingale. Conversely, if M is an integrable Markov martingale, $\mathcal{R}_{0,t} > 0$ is deterministic and multiplicative, and $X_t = m + \mathcal{R}_{0,t}M_t$, then X satisfies (24). The two processes have the same pair-copulas. If H_t is the marginal CDF of M_t , their quantiles satisfy

$$H_t^{-1}(u) = \frac{Q_t(u) - m}{\mathcal{R}_{0,t}}. \quad (28)$$

When the hypotheses of Theorem 2.8 hold, this is equivalent to

$$(\partial_t + \mathcal{A}_t)Q(t, u) - \gamma(t)\{Q_t(u) - m\} = 0. \quad (29)$$

Proof. Multiplicativity gives $\mathcal{R}_{0,t} = \mathcal{R}_{0,s}\mathcal{R}_{s,t}$. Hence

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = \frac{\mathcal{R}_{s,t}(X_s - m)}{\mathcal{R}_{0,t}} = M_s.$$

The converse follows by reversing this calculation. At each time the map between X_t and M_t is strictly increasing and affine, so it preserves pair-copulas and gives (28). Applying the generator martingale theorem to that quantile and differentiating $\mathcal{R}_{0,t}$ gives (29). $\square$

3.2 Exact copula and generator characterizations

Theorem 3.8 (Exact copula criterion for strong mean reversion). *Under Assumption 2.4, X satisfies (24) if and only if*

$$\int_I \{Q_t(v) - m\} c_{s,t}(u, v) dv = \mathcal{R}_{s,t}\{Q_s(u) - m\} \quad (30)$$

for every $s < t$ and Lebesgue-almost every u .

Proof. By (9), the left side of (30), evaluated at $u = U_s$, equals $\mathbb{E}[X_t - m \mid \mathcal{F}_s]$. Its right side then equals $\mathcal{R}_{s,t}(X_s - m)$. Since U_s is uniform, equality almost surely is equivalent to the displayed density identity for Lebesgue-almost every u . $\square$

Proposition 3.9 (Generator sign criterion). *Suppose $Q(t, u)$ is in the time-space extended generator domain and the local drift limit can be interchanged with the quantile representation. Then*

$$b_X(t, Q_t(u)) = (\partial_t + \mathcal{A}_t)Q(t, u). \quad (31)$$

Consequently, after separately imposing the limiting-mean condition, Definition 3.2 is equivalent to

$$\text{sgn}\{(\partial_t + \mathcal{A}_t)Q(t, u)\} = \text{sgn}\{m - Q_t(u)\} \quad (32)$$

for almost every interior u , with zero drift at $Q_t(u) = m$.

Proof. Insert $X_t = Q_t(U_t)$ into Definition 3.1 and use the definition of the time-space extended generator. Strict monotonicity converts the physical-state sign condition into (32). A complete kernel-integral proof, with separate bounded and weighted conditions for the moving time difference quotient, is given in Appendix A.2. $\square$

3.3 Stationary Gaussian copula and Gamma-margin diagnostics

Let Z be the stationary standardized OU process

$$dZ_t = -aZ_t dt + \sqrt{2a} dW_t, \quad a > 0, \quad Z_t \sim \mathcal{N}(0, 1). \quad (33)$$

Its copula is Gaussian with correlation $e^{-a(t-s)}$.

Proposition 3.10 (Stationary Gaussian-copula sign criterion). *Let F have a finite mean m , let $g(z) = F^{-1}(\Phi(z))$, and suppose g is strictly increasing, C^2 , and in the extended generator domain of (33). Let z_F be the unique number satisfying $g(z_F) = m$. If*

$$\text{sgn}\{g''(z) - zg'(z)\} = \text{sgn}(z_F - z) \quad \text{for every } z \in \mathbb{R}, \quad (34)$$

then $X_t = g(Z_t)$ is locally mean reverting toward m .

Proof. Itô's formula gives local drift $a\{g''(z) - zg'(z)\}$. Since g is strictly increasing, $\text{sgn}(m - g(z)) = \text{sgn}(z_F - z)$. Stationarity gives $\mathbb{E}[X_t] = m$ at every time. Integrability and the extended-domain assumption justify the drift calculation. $\square$

Example 3.11 (Stationary Gamma margin: a necessary root is not sufficient). Let G_ν be the quantile of $\text{Gamma}(\nu, 1)$, with $\nu > 0$, and set $Q(u) = sG_\nu(u)$, $s > 0$. Define

$$u_\nu = F_{\nu,1}(\nu), \quad z_\nu = \Phi^{-1}(u_\nu). \quad (35)$$

At the target mean $m = \nu s$, the zero-drift requirement reduces to

$$\Gamma(\nu) \left(\frac{e}{\nu}\right)^\nu \phi(z_\nu) - 2z_\nu = 0. \quad (36)$$

Equation (36) is only necessary. The model is locally mean reverting only if the full condition

$$\text{sgn}\{\mathcal{A}Q(u)\} = \text{sgn}\{\nu s - Q(u)\} \quad \text{for almost every } u \in I \quad (37)$$

also holds, with the required endpoint integrability. No existence claim is made from the scalar root alone.

Example 3.12 (Time-varying Gamma scale). Let $Q_t(u) = s(t)G_\nu(u)$, where $s(t) > 0$ is differentiable and $s(t) \rightarrow s_\infty \in (0, \infty)$. Write $z = \Phi^{-1}(u)$ and

$$\mathcal{G}_\nu(u) = 2\phi(z)zG'_\nu(u) - \phi(z)^2G''_\nu(u). \quad (38)$$

For the Gaussian-copula generator (17), the quantile drift is

$$D_t(u) = s'(t)G_\nu(u) - as(t)\mathcal{G}_\nu(u). \quad (39)$$

The correct target is $m = \nu s_\infty$, and the necessary and sufficient local sign test is

$$\operatorname{sgn} D_t(u) = \operatorname{sgn}\{m - s(t)G_\nu(u)\} \quad \text{for almost every } u \in I. \quad (40)$$

If u_t solves $s(t)G_\nu(u_t) = m$, then

$$\frac{s'(t)}{s(t)} = a \frac{\mathcal{G}_\nu(u_t)}{G_\nu(u_t)} \quad (41)$$

is a necessary zero-point equation, not a sufficient global condition. Its limit also inherits the stationary root requirement (36).

4 Mean-reverting monotone transforms of diffusions

4.1 Construction theorem and observable density

Let R be a Markov diffusion on an interval E ,

$$dR_t = b(R_t) dt + \sigma(R_t) dW_t, \quad \mathcal{L}h = bh_x + \frac{1}{2}\sigma^2 h_{xx}. \quad (42)$$

Theorem 4.1 (Strict diffusion-transform theorem). *Let $f \in C^{1,2}([0, \infty) \times E)$ and suppose that $x \mapsto f(t, x)$ is strictly increasing for every relevant t . Let γ be continuous and let*

$$\mathcal{R}_{s,t} = \exp\left\{\int_s^t \gamma(u) du\right\}. \quad (43)$$

Assume that

$$(\partial_t + \mathcal{L})f(t, x) = \gamma(t)\{f(t, x) - m\} \quad (44)$$

and, for every $T < \infty$,

$$\mathbb{E} \int_0^T \mathcal{R}_{0,t}^{-2} \sigma(R_t)^2 f_x(t, R_t)^2 dt < \infty. \quad (45)$$

Then $Y_t = f(t, R_t)$ is Markov and

$$\mathbb{E}[Y_t - m \mid \mathcal{F}_s] = \mathcal{R}_{s,t}(Y_s - m). \quad (46)$$

If $\gamma(t) < 0$, $\mathcal{R}_{s,t} \rightarrow 0$, and the initial first moment is finite, then Y has strong and local mean reversion toward m .

Conversely, if the strictly monotone $C^{1,2}$ transform satisfies (46) and its semimartingale decomposition is integrable, uniqueness of the drift part yields (44).

Proof. Itô's formula and (44) give

$$d(Y_t - m) = \gamma(t)(Y_t - m) dt + \sigma(R_t) f_x(t, R_t) dW_t.$$

Multiplying by $\mathcal{R}_{0,t}^{-1}$ removes the drift. Condition (45) makes the resulting stochastic integral a true square-integrable martingale on every finite horizon, proving (46). Strict invertibility transfers the Markov property from R to Y . The converse follows from uniqueness of the finite-variation part. $\square$

Proposition 4.2 (Transition density under a monotone transform). *Suppose $R_t \mid R_s = x_s$ has density $p_{s,t}(x_s, x_t)$ and the observations y_j lie in $f(j, E)$ with unique inverse states $x_j = f^{-1}(j, y_j)$ for $j = s, t$. Then*

$$q_{s,t}(y_s, y_t) = p_{s,t}(x_s, x_t) \left| \frac{\partial f^{-1}(t, y_t)}{\partial y_t} \right| = \frac{p_{s,t}(x_s, x_t)}{|f_x(t, x_t)|}. \quad (47)$$

This is a valid interior density only where the support, uniqueness, and Jacobian conditions hold. For the strictly increasing transforms used here, the probability ranks are unchanged; hence the transformed process has the same pair-copula density as the base process,

$$c_{s,t}^Y(u, v) = c_{s,t}^R(u, v). \quad (48)$$

Proof. Condition on the uniquely recovered state $R_s = x_s$ and apply the one-dimensional change-of-variables theorem at time t . $\square$

4.2 OU base process

Use the centered OU process

$$dR_t = -\kappa R_t dt + \sigma dW_t, \quad \kappa > 0, \quad \sigma > 0. \quad (49)$$

For $\Delta = t - s$,

$$R_t \mid R_s = x \sim \mathcal{N}\left(e^{-\kappa\Delta}x, \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa\Delta})\right). \quad (50)$$

The noncentered OU is obtained by replacing R_t with $R_t - \bar{r}$. It therefore satisfies

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}), \quad \lim_{t \rightarrow \infty} \mathbb{E}[R_t \mid \mathcal{F}_s] = \bar{r}. \quad (51)$$

Set $\gamma = -d$ with $d > 0$. The transform PDE is

$$f_t + \frac{1}{2}\sigma^2 f_{xx} - \kappa x f_x + d(f - m) = 0. \quad (52)$$

Example 4.3 (Affine transformed OU). For $A_0 > 0$ and $B_0 \in \mathbb{R}$,

$$f(t, x) = m + A_0 e^{(\kappa-d)t} x + B_0 e^{-dt} \quad (53)$$

solves (52) and is strictly increasing on $\mathbb{R}$. If the base OU is stationary, then

$$Y_t \sim \mathcal{N}\left(m + B_0 e^{-dt}, \frac{A_0^2 \sigma^2}{2\kappa} e^{2(\kappa-d)t}\right). \quad (54)$$

The variance may grow when $\kappa > d$; this does not contradict the first-moment identity (46), but the transformed process is then not stationary.

Proof. Substitution leaves the coefficient equations $A' + (d - \kappa)A = 0$ and $B' + dB = 0$. The derivative $f_x = A_0 e^{(\kappa-d)t}$ is positive. The marginal law follows from the affine Normal transformation. $\square$

Example 4.4 (Exponential transformed OU). Let

$$f(t, x) = m + e^{A(t)x+B(t)} + C(t), \quad (55)$$

$$A(t) = A_0 e^{\kappa t}, \quad (56)$$

$$B(t) = B_0 - \frac{\sigma^2 A_0^2}{4\kappa} (e^{2\kappa t} - 1) - dt, \quad (57)$$

$$C(t) = C_0 e^{-dt}. \quad (58)$$

If $A_0 > 0$, this is strictly increasing and its support is $(m + C(t), \infty)$. It is positive-valued when $m + C(t) \geq 0$. For stationary R , the shifted variable $Y_t - m - C(t)$ is lognormal with log mean $B(t)$ and log variance $A(t)^2 \sigma^2 / (2\kappa)$.

Proof. Substitution in (52) gives

$$A' = \kappa A, \quad B' = -\frac{1}{2}\sigma^2 A^2 - d, \quad C' = -dC,$$

whose solution is (56)–(58). The derivative $f_x = Ae^{Ax+B}$ is positive. Gaussian exponential moments verify (45) on finite horizons. $\square$

Proposition 4.5 (OU Kummer candidate is not an admissible monotone model). *Let $J(p, q, z)$ solve $zJ_{zz} + (q - z)J_z - pJ = 0$. The separated family*

$$f(t, x) = m + e^{-\zeta t} J\left(\frac{\zeta - d}{2\kappa}, \frac{1}{2}, \frac{\kappa x^2}{\sigma^2}\right) \quad (59)$$

solves (52) wherever it is smooth. Every nonconstant member is, however, noninjective on the OU state space $\mathbb{R}$ and therefore does not satisfy Theorem 4.1 or Proposition 4.2.

Proof. Direct substitution gives the stated Kummer equation. The spatial dependence is through x^2 , so $f(t, x) = f(t, -x)$ for every x . A nonconstant even function cannot be one-to-one on $\mathbb{R}$. $\square$

Remark 4.6. A density formed by summing two inverse branches would define a different observation model. The transformation would no longer preserve the base copula, and the observed scalar process need not be Markov; this is outside the present one-dimensional monotone framework.

4.3 CIR base process

Let

$$dR_t = \kappa(\bar{r} - R_t)dt + \sigma\sqrt{R_t}dW_t, \quad \kappa > 0, \quad \bar{r} > 0, \quad \sigma > 0, \quad (60)$$

with state space $[0, \infty)$. The Feller condition $2\kappa\bar{r} \geq \sigma^2$ is a convenient sufficient condition for zero to be unattainable when the initial state is positive. For $\Delta = t - s$,

$$R_t \mid R_s = x \stackrel{d}{=} c_\Delta \chi_\nu'^2(\lambda_\Delta), \quad (61)$$

$$c_\Delta = \frac{\sigma^2(1 - e^{-\kappa\Delta})}{4\kappa}, \quad \nu = \frac{4\kappa\bar{r}}{\sigma^2}, \quad \lambda_\Delta = \frac{4\kappa e^{-\kappa\Delta}x}{\sigma^2(1 - e^{-\kappa\Delta})}. \quad (62)$$

The conditional mean has the same linear reversion form as OU:

$$\mathbb{E}[R_t - \bar{r} \mid \mathcal{F}_s] = e^{-\kappa(t-s)}(R_s - \bar{r}). \quad (63)$$

When started from its invariant law,

$$R_t \sim \text{Gamma}\left(\frac{2\kappa\bar{r}}{\sigma^2}, \frac{\sigma^2}{2\kappa}\right), \quad (64)$$

where the second parameter is scale. With $\gamma = -d$, the corrected transform PDE is

$$f_t + \frac{1}{2}\sigma^2 x f_{xx} + \kappa(\bar{r} - x)f_x + d(f - m) = 0. \quad (65)$$

Example 4.7 (Affine transformed CIR). For $A_0 > 0$ and $B_0 \in \mathbb{R}$, define

$$A(t) = A_0 e^{(\kappa-d)t}, \quad (66)$$

$$B(t) = m - A_0 \bar{r} e^{(\kappa-d)t} + B_0 e^{-dt}, \quad (67)$$

$$f(t, x) = A(t)x + B(t). \quad (68)$$

Then f solves (65), is strictly increasing on the CIR state space, and has support $[B(t), \infty)$. The original manuscript's formula is the special case $m = \bar{r}$ after a harmless reparameterization of B_0 .

Proof. The affine coefficient equations are $A' + (d - \kappa)A = 0$ and $B' + d(B - m) + \kappa\bar{r}A = 0$. Equations (66)–(67) solve them, and $f_x = A > 0$. Polynomial moments of CIR verify the finite-horizon martingale condition. $\square$

Proposition 4.8 (Quadratic CIR transform: exact finite-horizon condition). *Let*

$$f(t, x) = A(t)x^2 + B(t)x + C(t), \quad A_0 > 0, \quad (69)$$

and set

$$\eta = \frac{A_0(\sigma^2 + 2\kappa\bar{r})}{\kappa}. \quad (70)$$

The coefficient solution is

$$A(t) = A_0 e^{(2\kappa - d)t}, \quad (71)$$

$$B(t) = e^{(\kappa - d)t} \{B_0 - \eta(e^{\kappa t} - 1)\}, \quad (72)$$

$$C(t) = m + \frac{\bar{r}\eta}{2} e^{(2\kappa - d)t} - \bar{r}(B_0 + \eta)e^{(\kappa - d)t} + \left(C_0 - m + \bar{r}B_0 + \frac{\bar{r}\eta}{2}\right) e^{-dt}. \quad (73)$$

It solves (65). On a fixed window $[0, T]$, the map $x \mapsto f(t, x)$ is increasing on $[0, \infty)$ for every $t \leq T$ if and only if

$$B_0 \geq \eta(e^{\kappa T} - 1). \quad (74)$$

Under a strict inequality the interior inverse Jacobian is strictly finite. No finite B_0 makes a nondegenerate quadratic transform globally increasing for all $t \geq 0$ under the standard CIR parameter restrictions.

Proof. Matching the coefficients of $x^2, x, 1$ gives

$$\begin{aligned} A' + (d - 2\kappa)A &= 0, \\ B' + (d - \kappa)B + (\sigma^2 + 2\kappa\bar{r})A &= 0, \\ C' + d(C - m) + \kappa\bar{r}B &= 0, \end{aligned}$$

which yields (71)–(73). Since $f_x = 2A(t)x + B(t)$ and $A(t) > 0$, monotonicity on the nonnegative half-line is equivalent to $B(t) \geq 0$. Formula (72) is nonnegative for all $t \leq T$ exactly under (74). Its bracket tends to $-\infty$ as $t \rightarrow \infty$, proving the global impossibility. $\square$

Remark 4.9 (Why $\kappa < 0$ is not the same CIR model). Algebraically, when $\kappa < 0$ the quadratic map can be globally increasing if $A_0 > 0$ and $\min\{B_0, B_0 + \eta\} \geq 0$. This does not rescue the standard CIR model. If $\bar{r} > 0$, the boundary drift $\kappa\bar{r} < 0$ points out of the nonnegative state space. If $\bar{r} < 0$ is chosen to make the intercept positive, the drift is explosive and there is no invariant Gamma law or mean reversion. Such a process would be a separately defined supercritical square-root diffusion, not a relaxation of (60).

Corollary 4.10 (Quadratic inverse and density). *Under the strict finite-window condition and for an interior observation $y > C(t)$, the unique nonnegative inverse is*

$$x = \frac{2\{y - C(t)\}}{B(t) + \sqrt{B(t)^2 + 4A(t)\{y - C(t)\}}}, \quad \left| \frac{\partial x}{\partial y} \right| = \frac{1}{2A(t)x + B(t)}. \quad (75)$$

Together with (61), this gives the finite-horizon observation density through (47).

Example 4.11 (Exponential transformed CIR). Let

$$f(t, x) = m + e^{A(t)x+B(t)} + D(t), \quad (76)$$

$$A(t) = \frac{2\kappa}{\sigma^2 + (2\kappa/A_0 - \sigma^2)e^{-\kappa t}}, \quad (77)$$

$$B(t) = -dt - \frac{2\kappa\bar{r}}{\sigma^2} \log[(2\kappa/A_0 - \sigma^2) + \sigma^2 e^{\kappa t}] + B_0 + \frac{2\kappa\bar{r}}{\sigma^2} \log(2\kappa/A_0), \quad (78)$$

$$D(t) = D_0 e^{-dt}. \quad (79)$$

For $A_0 > 0$, the logarithm arguments and denominator are positive and the map is strictly increasing. Its support is

$$[m + D(t) + e^{B(t)}, \infty). \quad (80)$$

If the base CIR is stationary, a sufficient and necessary initial exponential moment restriction is

$$0 < A_0 < \frac{2\kappa}{\sigma^2}. \quad (81)$$

Under these conditions the positive local-martingale component has constant expectation and is therefore a true martingale; this verifies the conditional mean identity even when the square-integrability criterion (45) is stronger than needed.

Proof. The coefficient equations are

$$A' + \frac{1}{2}\sigma^2 A^2 - \kappa A = 0, \quad B' + d + \kappa\bar{r}A = 0, \quad D' + dD = 0.$$

Their solution is (77)–(79). Since $R_t \geq 0$, the minimum of the exponential term is $e^{B(t)}$, proving (80). The stationary Gamma MGF is finite exactly below rate $2\kappa/\sigma^2$, which yields (81). More explicitly, put $p_0 = \sigma^2 A_0/(2\kappa) \in (0, 1)$ and $q = 2\kappa\bar{r}/\sigma^2$. Then

$$B(t) = B_0 - dt - q \log(1 - p_0 + p_0 e^{\kappa t}), \quad 1 - \frac{\sigma^2 A(t)}{2\kappa} = \frac{1 - p_0}{1 - p_0 + p_0 e^{\kappa t}}.$$

Using the stationary Gamma MGF shows

$$\mathbb{E} \exp\{dt + A(t)R_t + B(t)\} = e^{B_0}(1 - p_0)^{-q},$$

which is constant. The nonnegative local-martingale component is therefore a true martingale. $\square$

4.4 CIR special functions and the corrected reduction

Let $M(p, q, z) = {}_1F_1(p; q; z)$ denote Kummer's function and $U(p, q, z)$ Tricomi's function, with limiting definitions used at exceptional integer parameters.

Proposition 4.12 (CIR Kummer separated family). *Set*

$$p = \frac{\zeta - d}{\kappa}, \quad q = \frac{2\kappa\bar{r}}{\sigma^2}, \quad z = \frac{2\kappa x}{\sigma^2}. \quad (82)$$

If

$$J(p, q, z) = C_1 M(p, q, z) + C_2 U(p, q, z) \quad (83)$$

is a well-defined branch satisfying the relevant boundary conditions, then

$$f(t, x) = m + e^{-\zeta t} J(p, q, z) \quad (84)$$

solves (65). It defines a model in the present framework only if the selected branch is strictly increasing on $[0, \infty)$, has a unique stable inverse on its image, and satisfies the moment condition in Theorem 4.1 for the stated initial law.

For example, $p, q > 0$, $C_1 > 0$, $C_2 = 0$ gives

$$\partial_z M(p, q, z) = \frac{p}{q} M(p+1, q+1, z) > 0, \quad (85)$$

but this branch is not integrable under the stationary Gamma law because of its e^z asymptotic term. It may be integrable at each finite time from a deterministic initial state. Thus the separated equation is correct, whereas an unspecified “special-function CIR model” is not a complete statistical model.

Proof. Substitution gives $zJ_{zz} + (q-z)J_z - pJ = 0$. The derivative identity and standard positive series for M prove monotonicity in the stated branch. Its large- z behavior is $M(p, q, z) \sim \Gamma(q)e^z z^{p-q}/\Gamma(p)$; multiplying by the invariant $z^{q-1}e^{-z}$ density leaves a nonintegrable z^{p-1} tail for $p > 0$. $\square$

Proposition 4.13 (The second CIR transform in the original manuscript is false). *The exponential change of variables claimed in the original manuscript does not transform (65) into $u_\tau = zu_{zz}$; direct substitution has a nonzero residual for generic parameters. A correct direct reduction is*

$$f(t, x) - m = e^{-dt} u(\tau, z), \quad z = xe^{\kappa t}, \quad \tau = \frac{\sigma^2}{2\kappa} (1 - e^{\kappa t}), \quad (86)$$

which yields

$$u_\tau = zu_{zz} + \frac{2\kappa\bar{r}}{\sigma^2} u_z. \quad (87)$$

Therefore the three functions listed after the false reduction are not thereby solutions of the original CIR transform PDE.

Proof. Differentiate (86) and substitute into (65); the κzu_z terms cancel and the remaining terms give (87). The coefficient of u_z is strictly positive under the standard CIR parameters and cannot generally be discarded. $\square$

5 Mixed copula families as stationary resets

5.1 Transition-density construction and preservation of mean reversion

Let $P_{s,t}$ be a Markov evolution on a state space E with common invariant probability law π . Assume that, for $s < t$, both are dominated by a common σ -finite measure μ_0 ,

$$P_{s,t}(x, dy) = p_{s,t}(x, y) \mu_0(dy), \quad \pi(dy) = p_\pi(y) \mu_0(dy).$$

Define the independence/reset operator

$$\Pi(x, A) = \pi(A), \quad (\Pi h)(x) = \int_E h(z) \pi(dz). \quad (88)$$

Let $\lambda : [0, \infty) \rightarrow [0, \infty)$ be right-continuous and locally bounded, and set

$$a_{s,t} = \exp \left\{ - \int_s^t \lambda(u) du \right\}. \quad (89)$$

Theorem 5.1 (Consistent reset density mixture). *The kernels*

$$P_{s,t}^{\text{mix}} = a_{s,t}P_{s,t} + (1 - a_{s,t})\Pi \quad (90)$$

form a Markov evolution and preserve π . Their transition densities are

$$p_{s,t}^{\text{mix}}(x, y) = a_{s,t}p_{s,t}(x, y) + (1 - a_{s,t})p_{\pi}(y). \quad (91)$$

Assume additionally that $E \subseteq \mathbb{R}$, the process is started from π , the invariant CDF F_{π} is continuous and strictly increasing on the interior of its support, and the stationary base pair-copula has density $c_{s,t}$. Then the mixed pair-copula and its density are

$$C_{s,t}^{\text{mix}} = a_{s,t}C_{s,t} + (1 - a_{s,t})\Pi_C, \quad \Pi_C(u, v) = uv, \quad (92)$$

$$c_{s,t}^{\text{mix}}(u, v) = a_{s,t}c_{s,t}(u, v) + (1 - a_{s,t}). \quad (93)$$

If the base generator is $\mathcal{G}_t$, then, on any invariant generator domain whose functions are integrable under π ,

$$\mathcal{G}_t^{\text{mix}}h(x) = \mathcal{G}_th(x) + \lambda(t) \left\{ \int_E h(z)\pi(dz) - h(x) \right\}. \quad (94)$$

Proof. The survival weights are multiplicative. Invariance gives $P\Pi = \Pi P = \Pi$ and $\Pi^2 = \Pi$. Expanding $P_{s,r}^{\text{mix}}P_{r,t}^{\text{mix}}$ therefore reduces it to (90) with weight $a_{s,r}a_{r,t} = a_{s,t}$. Taking densities gives (91); uniformization under stationary start gives (92) and (93). Its margins equal one, and direct expansion using (2)–(3) gives

$$\int_I c_{s,r}^{\text{mix}}(u, w)c_{r,t}^{\text{mix}}(w, v)dw = a_{s,t}c_{s,t}(u, v) + (1 - a_{s,t}) = c_{s,t}^{\text{mix}}(u, v).$$

The right derivative of (90) at $t = s+$ gives (94). $\square$

Theorem 5.2 (Exact strengthening of conditional mean reversion). *Suppose the base process has invariant mean $m = \int_E z\pi(dz)$ and satisfies*

$$\mathbb{E}[Z_t - m \mid Z_s = x] = \mathcal{R}_{s,t}(x - m). \quad (95)$$

Then the reset-mixture process satisfies

$$\mathbb{E}[Y_t - m \mid Y_s = x] = a_{s,t}\mathcal{R}_{s,t}(x - m). \quad (96)$$

Its local drift is

$$b_{\text{mix}}(t, x) = b_0(t, x) + \lambda(t)(m - x). \quad (97)$$

Thus a nonnegative reset intensity preserves the local sign condition and, for $\lambda > 0$, strictly accelerates the decay of conditional mean deviations away from m .

Proof. The base component of (90) contributes $a_{s,t}\mathcal{R}_{s,t}(x - m)$. A draw from π has centered mean zero, so the reset component contributes zero. This proves (96); its right derivative gives (97). $\square$

Remark 5.3 (A precise theoretical advantage). The mixed model does not alter any one-time marginal under stationary start, but multiplies the base conditional-mean persistence by $a_{s,t}$. Hence $a_{s,t}\mathcal{R}_{s,t}$, the implied half-life, or the additive rate $-\partial_{t+} \log a_{s,t} = \lambda(t)$ are clear structural measures of its extra reversion channel. Whether this produces superior out-of-sample log scores or calibration is a separate empirical question.

5.2 Poisson realization

Theorem 5.4 (Independent-copy reset construction). *Let $\{Z^{(j)} : j \geq 0\}$ be independent copies of a stationary Markov process with transition evolution P and invariant law π . Let N be an independent nonhomogeneous Poisson process with intensity $\lambda(t)$, and set*

$$Y_t = Z_t^{(N_t)}. \quad (98)$$

Then Y is Markov, stationary in its one-time marginals, and has transition kernel (90) and transition density (91). Under the stationary continuous-margin conditions of Theorem 5.1, it also has Copula density (93). A constant λ gives $a_{s,t} = e^{-\lambda(t-s)}$; a positive deterministic seasonal intensity is also admissible.

Proof. Conditional on no jump in $(s, t]$, whose probability is $a_{s,t}$, the same copy evolves with kernel $P_{s,t}$. Conditional on at least one jump, the active copy at t is independent of the history through time s and has law π . The resulting conditional law is (90), whose density is (91), and it is a Markov evolution by Theorem 5.1. $\square$

6 Risk-neutral bond pricing with reset risk

This section is a pricing theorem under a risk-neutral measure $\mathbb{Q}$, not a claim that a physical-measure stablecoin spot series identifies bond prices.

Theorem 6.1 (Nonlocal Feynman–Kac equation). *Under $\mathbb{Q}$, let the short rate be a stationary-reset OU process with generator*

$$\mathcal{G}h(y) = \kappa(m - y)h'(y) + \frac{1}{2}\sigma^2 h''(y) + \lambda \left\{ \int_{\mathbb{R}} h(z)\pi(dz) - h(y) \right\}, \quad (99)$$

where $\pi = \mathcal{N}(m, \sigma^2/(2\kappa))$. Suppose the exponential discount functional is integrable and that the classical Feynman–Kac hypotheses hold; alternatively, assume conditions giving a unique viscosity solution with the stated growth class. Then

$$g(t, y) = \mathbb{E}^{\mathbb{Q}} \left[\exp \left\{ - \int_t^T Y_s ds \right\} \middle| Y_t = y \right] \quad (100)$$

solves

$$\begin{cases} g_t + \kappa(m - y)g_y + \frac{1}{2}\sigma^2 g_{yy} + \lambda \left\{ \int_{\mathbb{R}} g(t, z)\pi(dz) - g(t, y) \right\} - yg = 0, \\ g(T, y) = 1. \end{cases} \quad (101)$$

Proof. Theorem 5.1 gives the generator (99). Apply the Feynman–Kac formula with killing rate y and terminal payoff one. Rearranging the generator equation yields (101). $\square$

7 Embedding of the empirical specifications

The following model identities are part of the theory–empirics interface. They prevent four distinct notions—a coded forward map, an invertible map, a normalized transition density, and a successfully estimated structural model—from being collapsed into one “implemented” flag.

Use the following classes:

- S** one-dimensional structural model covered by the corrected theorems;
- C** covered only after an explicit condition, a finite horizon, or a change in the observed state;
- L** legitimate latent-state or horizon-specific predictive extension, but not a one-dimensional continuous-time Markov model for the observation;
- B** benchmark not carrying the structural claim;

N not embeddable under its current model identity.

No.	Empirical registry name	Class	Correct theoretical interpretation
1	OUF	S	Exact OU transition; $\kappa, \sigma > 0$ and finite first moment give strong reversion.
2	ThresholdOU_ Heteroskedastic	B	The drift points to a threshold, but with unequal $\kappa_{\pm}$ the invariant mean is not that threshold. The current Euler Gaussian QMLE is not the exact threshold-diffusion transition.
3	CIR_Price	S	Standard CIR in theory. Any Normal moment-matching numerical fallback must be disclosed as an approximate likelihood rather than an exact CIR density.
4	CIR_DepegPressure	C	Valid only when the absolute depeg amplitude itself is defined as the CIR state. The sign of the price deviation is lost; a signed-price model requires an augmented sign state.
5	AffineTransformedOU	S	Example 4.3 with a unique inverse and exact Jacobian.
6	Exponential TransformedOU	C	Example 4.4, conditional on support and the intended target m ; the original zero target must not be interpreted automatically as one dollar.
7	SpecialFunction TransformedOU	N	Proposition 4.5; a forward map and derivative do not constitute an inverse transition density.
8	AffineTransformedCIR	S	Example 4.7, subject to CIR boundary and observation-support checks.
9	Quadratic TransformedCIR	C	Proposition 4.8; only a stated finite window satisfying (74).
10	Exponential TransformedCIR	C	Example 4.11; enforce (81) under stationary start and the exact lower support (80).
11	SpecialFunction TransformedCIR	C	Proposition 4.12; a specified branch must pass monotonicity, inverse, support, moment, and numerical-stability checks. A finite branch scan is diagnostic, not a full density implementation.
12	MOUF	S	Constant-intensity stationary OU reset; Theorems 5.1–5.4.
13	MCIR	S	Stationary CIR reset under the full-density and no-boundary-atom conditions of Proposition 2.10; a base-density fallback and weak identification are empirical qualifications, not changes to the density-mixture theorem.
14	SeasonalIntensityMOU	S	The nonhomogeneous Poisson version of Theorem 5.4, provided the nonnegative seasonal intensity is right-continuous and locally bounded.
15	JumpOU	B	An exact compound-Poisson OU with $\sigma > 0$ is an absolutely continuous jump-diffusion extension; the pure-jump boundary $\sigma = 0$ has a no-jump atom and lies outside the present Copula-density scope. With jump mean μ_J , its mean target is $\bar{r} + \lambda\mu_J/\kappa$. The current one-jump moment approximation is not a proved Chapman–Kolmogorov semigroup.
16	MarkovSwitchingOU	L	(X_t, S_t) is Markov; X_t alone generally is not. Structural claims must be stated on the augmented regime state.
17	NestedOU	L	The vector of two OU factors is Markov. Its scalar sum is Markov only in degenerate/equal-speed cases; pairwise Gaussian composite scores are not a full Kalman path likelihood.
18	RandomWalk	B	Explicit nonreverting negative control.
19	GaussianGamma Stationary	N	It is a valid stationary Markov density, but it is not a proved mean-reverting model until (37) passes; the scalar root is insufficient.
20	GaussianGamma TimeVarying	N	The endpoint ODE (41) is only diagnostic. A scale path, full density, and global sign test (40) are still required.
21	ParametricCopulaGrid	L	Gaussian $\rho(h) = e^{-\kappa h}$ and its independence-reset mixture have continuous-time semigroups. At admissible interior parameters, Student- t , Clayton, Gumbel, and Frank cells give absolutely continuous one-step Copula densities but lack a proved same-family continuous-time embedding. Singular boundary parameters are excluded. Iterating a fixed five-minute Copula-density transition operator defines a discrete-time Markov chain, not the claimed parametric semigroup.
22	TwoScaleGaussian	L	A two-OU latent vector is valid; the scalar correlation mixture is generally non-Markov and lies outside the one-dimensional generator theorem.
23	TwoScaleMixed	L	A rigorous model resets the entire latent vector at Poisson times. Pairwise scalar mixed densities alone do not prove the density Chapman–Kolmogorov identity or process-level consistency.
24	FullyTimeVarying MarginalCopula	C	If $X_t = \mu(t) + s(t)Z_t$ with standardized OU Z and deterministic $s(t) > 0$, then X itself is a one-dimensional time-inhomogeneous Markov process with exact Gaussian transition and Copula densities. Reversion to fixed m additionally requires $\beta(t) = \kappa - s'(t)/s(t) > 0$, $\mu'(t) = \beta(t)\{m - \mu(t)\}$, and $\mu(t) \rightarrow m$. Permanent seasonality or a nonzero linear trend violates the limiting-mean condition, not Markovity.
25	MixedCopulaBondPDE	S	Theorem 6.1 under $\mathbb{Q}$. It is not identifiable from a physical-measure USDT spot series without bond-price data.

7.1 Further formulas for the extension classes

For a two-sided threshold diffusion centered at c , with drift $-\kappa_+(x-c)$ above c and $-\kappa_-(x-c)$ below c , and piecewise constant volatilities $\sigma_\pm$, its invariant density is proportional to

$$\pi(x) \propto \begin{cases} \sigma_+^{-2} \exp\{-\kappa_+(x-c)^2/\sigma_+^2\}, & x \geq c, \\ \sigma_-^{-2} \exp\{-\kappa_-(x-c)^2/\sigma_-^2\}, & x < c. \end{cases} \quad (102)$$

Direct integration shows $\int (x-c)\pi(x)dx = 0$ if and only if $\kappa_+ = \kappa_-$. Thus directional drift toward c is not enough to call c the asymptotic mean.

For a time-varying Gaussian-margin model $X_t = \mu(t) + s(t)Z_t$, where $dZ_t = -\kappa Z_t dt + \sqrt{2\kappa} dW_t$, Itô's formula gives

$$b_X(t, x) = \mu'(t) + \left\{ \frac{s'(t)}{s(t)} - \kappa \right\} \{x - \mu(t)\}. \quad (103)$$

Writing $\beta(t) = \kappa - s'(t)/s(t)$, this equals $-\beta(t)(x - m)$ precisely when

$$\beta(t) > 0, \quad \mu'(t) = \beta(t)\{m - \mu(t)\}. \quad (104)$$

For an OU jump diffusion with compound-Poisson jump mean μ_J and intensity λ , the forward drift is

$$\kappa(\bar{r} - x) + \lambda\mu_J = \kappa \left(\bar{r} + \frac{\lambda\mu_J}{\kappa} - x \right). \quad (105)$$

Hence using the base OU level as the target is valid only for centered jumps.

8 What the corrected theory does and does not claim

- (i) Exact finite-time identities such as (30) take priority over a formal local PDE. A zero generator drift is not silently upgraded from local to true martingale.
- (ii) A formal PDE solution becomes an observable one-dimensional Markov model only after strict monotonicity, support, inverse-Jacobian, moment, and boundary conditions pass.
- (iii) The stationary and time-varying Gamma equations are retained as necessary diagnostics, not existence theorems.
- (iv) The quadratic CIR transform is a finite-horizon model. The OU even special-function transform is excluded from the monotone framework. The CIR Kummer family is branch-specific, and the erroneous second CIR reduction is replaced by (87).
- (v) The absolutely continuous mixed-copula reset construction is rigorous and gives the exact structural improvement (96); empirical superiority still has to be demonstrated by common out-of-sample density and calibration metrics.
- (vi) Hidden-regime, two-factor, and two-scale specifications belong to an augmented-state theory. Horizon-specific copula fits and transition approximations are prediction models unless a consistent continuous-time semigroup is separately proved.

A Detailed proofs

This appendix expands the measure-theoretic steps that are easy to hide inside phrases such as “interchange the limit and the conditional expectation.” It covers the general results used repeatedly in the main text. The model-specific OU, CIR, and special-function calculations remain next to their statements because they are direct substitutions into the displayed differential equations.

A.1 Short-time kernels acting on moving functions

Lemma A.1 (Moving-function kernel lemma). *Fix $t \geq 0$ and an interior state $u \in I$. Write*

$$K_h(u, dv) = c_{t,t+h}(u, v) dv, \quad h > 0,$$

so that $K_h(u, I) = 1$. The following statements hold.

(a) *Suppose that*

$$K_h(u, \{v : |v - u| \geq \delta\}) \longrightarrow 0 \quad \text{for every } \delta > 0. \quad (106)$$

If g is bounded and continuous at u , then

$$\int_I g(v) K_h(u, dv) \longrightarrow g(u). \quad (107)$$

(b) *In addition to part (a), let $r_h : I \rightarrow \mathbb{R}$ be measurable and suppose g is bounded and continuous at u and*

$$\|r_h - g\|_\infty \longrightarrow 0. \quad (108)$$

Then

$$\int_I r_h(v) K_h(u, dv) \longrightarrow g(u). \quad (109)$$

(c) *For an unbounded g , let $w : I \rightarrow [1, \infty)$ be measurable and define $\|f\|_w = \sup_{v \in I} |f(v)|/w(v)$. If*

$$\|r_h - g\|_w \longrightarrow 0, \quad (110)$$

$$\sup_{0 < h < h_0} \int_I w(v) K_h(u, dv) < \infty, \quad (111)$$

$$\int_I g(v) K_h(u, dv) \longrightarrow g(u), \quad (112)$$

then (109) still holds.

Proof. For part (a), split the integral at distance δ from u :

$$\begin{aligned} \left| \int_I g(v) K_h(u, dv) - g(u) \right| &\leq \int_I |g(v) - g(u)| K_h(u, dv) \\ &\leq \sup_{|v-u| < \delta} |g(v) - g(u)| + 2\|g\|_\infty K_h(u, \{|v - u| \geq \delta\}). \end{aligned}$$

First let $h \downarrow 0$ and use (106); then let $\delta \downarrow 0$ and use continuity of g at u . This proves (107), or equivalently $K_h(u, \cdot) \Rightarrow \delta_u$.

For part (b), row normalization gives

$$\begin{aligned} \left| \int_I r_h(v) K_h(u, dv) - g(u) \right| &\leq \int_I |r_h(v) - g(v)| K_h(u, dv) \\ &\quad + \left| \int_I g(v) K_h(u, dv) - g(u) \right| \\ &\leq \|r_h - g\|_\infty + \left| \int_I g(v) K_h(u, dv) - g(u) \right|. \end{aligned}$$

Both terms tend to zero by (108) and part (a).

For part (c), replace the first term in the preceding bound by

$$\int_I |r_h(v) - g(v)| K_h(u, dv) \leq \|r_h - g\|_w \int_I w(v) K_h(u, dv).$$

Equations (110)–(112) prove the claim. This weighted version is the relevant one for Normal, Student, or other quantiles that diverge at the endpoints of I . $\square$

Pointwise convergence $r_h(v) \rightarrow g(v)$ alone is not enough in this lemma: the measure $K_h(u, dv)$ also changes with h . Conditions (108) or (110)–(111) provide the uniform integrability needed for the moving integrand.

A.2 Full integral proof of the quantile–generator drift identity

Detailed proof of Proposition 3.9. Fix t and put $q_t(v) = q(t, v) = Q_t(v)$ and $\dot{q}_t(v) = \partial_t q(t, v)$. Assume that all displayed first moments are finite and that, for the state u under consideration,

$$\mathcal{A}_t q_t(u) = \lim_{h \downarrow 0} \frac{\int_I q_t(v) c_{t,t+h}(u, v) dv - q_t(u)}{h} \quad (113)$$

exists. Also assume either part (b) or part (c) of Lemma A.1 with

$$r_h(v) = \frac{q(t+h, v) - q(t, v)}{h}, \quad g(v) = \dot{q}_t(v). \quad (114)$$

These are explicit sufficient conditions for the limit-interchange clause in Proposition 3.9.

For a regular conditional kernel version, strict invertibility of q_t gives, for Leb-almost every u ,

$$P_{t,t+h}^X(q_t(u), B) = \int_I \mathbf{1}_B(q_{t+h}(v)) c_{t,t+h}(u, v) dv. \quad (115)$$

Consequently,

$$b_X(t, q_t(u)) = \lim_{h \downarrow 0} \frac{\int_I q_{t+h}(v) c_{t,t+h}(u, v) dv - q_t(u)}{h}. \quad (116)$$

Using $\int_I c_{t,t+h}(u, v) dv = 1$, the difference quotient in (116) is exactly

$$\int_I \frac{q_{t+h}(v) - q_t(v)}{h} c_{t,t+h}(u, v) dv \quad (117)$$

$$+ \frac{\int_I q_t(v) c_{t,t+h}(u, v) dv - q_t(u)}{h}. \quad (118)$$

Lemma A.1 and (114) show that (117) tends to $\dot{q}_t(u)$. Definition (113) shows that (118) tends to $\mathcal{A}_t q_t(u)$. Hence

$$b_X(t, Q_t(u)) = \partial_t q(t, u) + \mathcal{A}_t q(t, \cdot)(u) = (\partial_t + \mathcal{A}_t) q(t, u). \quad (119)$$

Because U_t is uniform, the statement is naturally a Leb-almost-everywhere statement in u . A continuous kernel version extends it to every interior state where the limits exist. Strict monotonicity of q_t then transfers the sign of the right-hand side to the physical state $x = q_t(u)$, which proves the sign assertion in Proposition 3.9 after the limiting-mean condition is imposed separately. $\square$

A.3 Conditional expectations and observable transition densities

Detailed proof of Theorems 2.5 and 3.8, and Proposition 2.6. For $s < t$, define

$$H_{s,t}(u) = \int_I q_t(v) c_{s,t}(u, v) \, dv.$$

Column normalization and Tonelli's theorem give

$$\begin{aligned} \int_I |H_{s,t}(u)| \, du &\leq \int_I \int_I |q_t(v)| c_{s,t}(u, v) \, dv \, du \\ &= \int_I |q_t(v)| \left\{ \int_I c_{s,t}(u, v) \, du \right\} \, dv \\ &= \int_I |q_t(v)| \, dv < \infty. \end{aligned}$$

For a bounded $\mathcal{F}_s$ -measurable random variable Z , the Markov property, first for bounded truncations of q_t and then by L^1 convergence, gives

$$\begin{aligned} \mathbb{E}[ZX_t] &= \mathbb{E}[Zq_t(U_t)] \\ &= \mathbb{E}\left[Z \int_I q_t(v) c_{s,t}(U_s, v) \, dv\right] = \mathbb{E}[ZH_{s,t}(U_s)]. \end{aligned}$$

This is the defining test-function identity for

$$\mathbb{E}[X_t \mid \mathcal{F}_s] = H_{s,t}(U_s).$$

Since $X_s = q_s(U_s)$ and $U_s \sim \text{Leb}$, equality $H_{s,t}(U_s) = q_s(U_s)$ almost surely is equivalent to $H_{s,t}(u) = q_s(u)$ for Leb-almost every u . This proves both directions of Theorem 2.5. Replacing the last equality by

$$H_{s,t}(u) - m = \mathcal{R}_{s,t}\{q_s(u) - m\}$$

proves both directions of Theorem 3.8.

Under the density assumptions of Proposition 2.6, the substitution $v = F_t(y)$, $dv = p_t(y) \, dy$, in the conditional kernel gives

$$p_{s,t}^X(x, y) = c_{s,t}\{F_s(x), F_t(y)\} p_t(y).$$

The three required identities follow by explicit changes of variables:

$$\begin{aligned} \int_{E_t^\circ} p_{s,t}^X(x, y) \, dy &= \int_I c_{s,t}\{F_s(x), v\} \, dv = 1, \\ \int_{E_s^\circ} p_s(x) p_{s,t}^X(x, y) \, dx &= p_t(y) \int_I c_{s,t}\{u, F_t(y)\} \, du = p_t(y), \\ \int_{E_r^\circ} p_{s,r}^X(x, z) p_{r,t}^X(z, y) \, dz &= p_t(y) \int_I c_{s,r}\{F_s(x), w\} c_{r,t}\{w, F_t(y)\} \, dw \\ &= p_t(y) c_{s,t}\{F_s(x), F_t(y)\} = p_{s,t}^X(x, y). \end{aligned}$$

The final equality is precisely the density Chapman–Kolmogorov identity. $\square$

A.4 The generator martingale theorem and rescaling PDE

Detailed proof of Theorem 2.8. Write

$$B(t, u) = (\partial_t + \mathcal{A}_t)q(t, u).$$

Membership in the time–space extended generator domain means that, after a localizing sequence if necessary,

$$L_t = q(t, U_t) - q(0, U_0) - \int_0^t B(s, U_s) ds \quad (120)$$

is a local martingale and the finite-variation integral is locally finite. If $B = 0$ for $dt \otimes \text{Leb}(du)$ -almost every (t, u) , uniformity of U_s and Tonelli's theorem yield, for every $T < \infty$,

$$\mathbb{E} \int_0^T \mathbf{1}_{\{B(s, U_s) \neq 0\}} ds = \int_0^T \int_I \mathbf{1}_{\{B(s, u) \neq 0\}} du ds = 0.$$

Thus the integral in (120) vanishes and $q(t, U_t)$ is a local martingale. Class- (D) on each finite horizon makes the stopped family uniformly integrable, so the local martingale is a true martingale.

Conversely, suppose $q(t, U_t)$ is a true martingale. It is also a local martingale, and subtracting the local martingale L in (120) shows that

$$A_t := \int_0^t B(s, U_s) ds$$

is a local martingale. It is an absolutely continuous finite-variation process starting from zero. A finite-variation local martingale is indistinguishable from a constant, so $A_t = 0$ for all t almost surely and $B(s, U_s) = 0$ for $ds \otimes \mathbb{P}$ -almost every (s, ω) . Uniformity of U_s and Tonelli's theorem now give

$$0 = \int_0^T \mathbb{P}\{B(s, U_s) \neq 0\} ds = \int_0^T \int_I \mathbf{1}_{\{B(s, u) \neq 0\}} du ds.$$

Hence $B = 0$ for $dt \otimes \text{Leb}(du)$ -almost every (t, u) , proving the converse in Theorem 2.8. $\square$

Detailed generator proof of Proposition 3.7. For completeness, let $R_t = \mathcal{R}_{0,t} = \exp\{\int_0^t \gamma(a) da\}$ and define

$$\tilde{q}(t, u) = \frac{q(t, u) - m}{R_t}.$$

Since R_t is deterministic, $R'_t = \gamma(t)R_t$, and $\mathcal{A}_t$ acts only on the state variable,

$$(\partial_t + \mathcal{A}_t)\tilde{q}(t, u) = \frac{1}{R_t} [(\partial_t + \mathcal{A}_t)q(t, u) - \gamma(t)\{q(t, u) - m\}]. \quad (121)$$

Thus the rescaled process is a martingale, subject to the preceding domain and class- (D) qualifications, exactly when (29) holds. At the finite-horizon level, multiplicativity gives $R_t = R_s \mathcal{R}_{s,t}$ and therefore

$$\mathbb{E} \left[\frac{X_t - m}{R_t} \middle| \mathcal{F}_s \right] = \frac{\mathcal{R}_{s,t}(X_s - m)}{R_s \mathcal{R}_{s,t}} = \frac{X_s - m}{R_s}.$$

The reverse calculation proves the converse part of Proposition 3.7 without appealing only to formal PDE algebra. $\square$

A.5 Diffusion transforms and reset mixtures

Detailed proof of Theorem 4.1. For Theorem 4.1, Itô's formula gives

$$\begin{aligned} d\{f(t, R_t) - m\} &= (\partial_t + \mathcal{L})f(t, R_t) dt + \sigma(R_t)f_x(t, R_t) dW_t \\ &= \gamma(t)\{f(t, R_t) - m\} dt + \sigma(R_t)f_x(t, R_t) dW_t. \end{aligned}$$

With $R_t^\gamma = \exp\{\int_0^t \gamma(a) da\}$, the deterministic product rule yields

$$d\left[\frac{f(t, R_t) - m}{R_t^\gamma}\right] = \frac{\sigma(R_t)f_x(t, R_t)}{R_t^\gamma} dW_t. \quad (122)$$

Condition (45) makes the stochastic integral in (122) square integrable on every finite horizon. Taking the conditional expectation between s and t and using $R_t^\gamma/R_s^\gamma = \mathcal{R}_{s,t}$ gives (46). Strict monotonicity gives the time-dependent inverse $R_s = f^{-1}(s, Y_s)$ and the transition kernel

$$P_{s,t}^Y(y, B) = P_{s,t}^R(f^{-1}(s, y), f^{-1}(t, B)),$$

which proves the Markov claim. Conversely, the conditional-mean identity makes the left side of (122) a martingale; uniqueness of the finite-variation term in Itô's decomposition forces (44) wherever the coefficients and derivatives are defined. $\square$

Detailed proof of Theorems 5.1 and 5.2. Finally, write $a = a_{s,r}$ and $b = a_{r,t}$. Invariance gives $P_{s,r}\Pi = \Pi P_{r,t} = \Pi$ and $\Pi^2 = \Pi$, so the reset-kernel composition in Theorem 5.1 expands as

$$\begin{aligned} & \{aP_{s,r} + (1-a)\Pi\}\{bP_{r,t} + (1-b)\Pi\} \\ &= abP_{s,t} + \{a(1-b) + (1-a)b + (1-a)(1-b)\}\Pi \\ &= abP_{s,t} + (1-ab)\Pi = P_{s,t}^{\text{mix}}, \end{aligned}$$

because $ab = a_{s,t}$. For the Copula densities the corresponding integral is

$$\begin{aligned} & \int_I \{ac_{s,r}(u, w) + 1 - a\}\{bc_{r,t}(w, v) + 1 - b\} dw \\ &= abc_{s,t}(u, v) + a(1-b) + (1-a)b + (1-a)(1-b) \\ &= a_{s,t}c_{s,t}(u, v) + 1 - a_{s,t}, \end{aligned}$$

where row normalization, column normalization, and the base Chapman–Kolmogorov identity were used term by term. Moreover, $a_{t,t+h} = 1 - \lambda(t)h + o(h)$ and $P_{t,t+h}h_0 = h_0 + h\mathcal{G}_th_0 + o(h)$ on the generator domain. Therefore

$$\frac{P_{t,t+h}^{\text{mix}}h_0 - h_0}{h} = \mathcal{G}_th_0 + \lambda(t)\{\Pi h_0 - h_0\} + o(1),$$

which proves (94). The conditional centered first moment is, directly in integral form,

$$\begin{aligned} \int_E (y - m)P_{s,t}^{\text{mix}}(x, dy) &= a_{s,t} \int_E (y - m)P_{s,t}(x, dy) \\ &\quad + (1 - a_{s,t}) \int_E (y - m)\pi(dy) \\ &= a_{s,t}\mathcal{R}_{s,t}(x - m), \end{aligned}$$

and its right derivative at $t = s$ gives $b_{\text{mix}}(s, x) = b_0(s, x) + \lambda(s)(m - x)$. $\square$